Financial Econometric Class

Course Syllabus, Foundations, Asset Pricing, and Time Series Analysis

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Course Overview

1. Syllabus & Assessment
2. Research Traditions & Foundations
3. Classical Linear Regression Model
4. Asset Pricing Models
5. Univariate Time Series Models
6. Summary & Next Steps

Course Description & Objectives

Course Philosophy

This course introduces students to empirical financial research methodology, bridging financial theory and econometric tools. It prepares students to critically evaluate, replicate, and produce high-quality empirical studies in finance.

Core Learning Outcomes

- Understand the epistemology and traditions of modern financial research.
- Master the Classical Linear Regression Model (CLRM) and diagnostic testing.
- Implement classical and modern Asset Pricing models (CAPM, Fama-French).
- Model and forecast univariate financial time series data.

Assessment Scheme & Grading Structure

Grading Components (Bobot Penilaian)

The course evaluation is divided across continuous coursework, empirical research papers, and written examinations.

Grading Weight Distribution

Component	**Description**	**Weight**
Continuous Assessment	Homework, Quizzes, Lab Work (R/EViews)	10%
Mid-term Paper	Written empirical exercise (Makalah Paruh 1)	20%
Mid-term Exam (UTS)	Written exam covering Sessions 1–4	25%
End-of-term Project	Group/Individual presentation & final paper	20%
Final Exam (UAS)	Comprehensive written exam	25%
Total		**100%**

Weekly Schedule & Software Tools

Weekly Topic Outline

1. **Week 1:** Introduction to Financial Research Concepts
2. **Week 2:** Theory & Empirical Models in Finance
3. **Week 3:** Classical Linear Regression Model (CLRM)
4. **Week 4:** Multiple Regression & Diagnostic Assumptions
5. **Week 5–6:** Asset Pricing Models (Fama-French 1992, 1996)
6. **Week 7:** Univariate Time Series Concepts & ARIMA

Lab Software & Data Databases

- **Econometric Software:** R (packages: `tseries`, `moments`), EViews, Stata.
- **Data Sources:** Refinitiv Eikon / Datastream, Capital IQ.

Evolution of Financial Research Paradigms

Three Historical Eras

- **Old Finance (1930s–1960s):** Qualitative, accounting/legal focus, Graham & Dodd security analysis.
- **Neoclassical / Modern Finance (1950s–1990s):** Grounded in rational agents, market efficiency (EMH), portfolio selection (Markowitz), CAPM, Modigliani-Miller.
- **New Finance (1990s–Present):** Relaxing rationality and efficiency; multi-factor models, behavioral finance, ESG, financial technology.

Neoclassical Pillars

Neoclassical finance rests on three core assumptions: (1) Rational agents, (2) Competitive / No-arbitrage markets, and (3) Perfect information.

Measurement Error & Research Validity

Research Validity Metrics

- **Internal Validity:** Degree to which causal claims are trustworthy (threatened by omitted variables, survivorship bias).
- **External Validity:** Generalizability across populations, periods, and markets.

Measurement Error & Reliability Ratio

A measured variable M consists of the true variable T plus error u:

$$M = T + u$$

The attenuation ratio or Reliability Ratio R measures signal-to-noise strength:

$$R = \frac{\mathrm{Var}(T)}{\mathrm{Var}(M)} = \frac{\mathrm{Var}(T)}{\mathrm{Var}(T) + \mathrm{Var}(u)} = \frac{1}{1 + \frac{\mathrm{Var}(u)}{\mathrm{Var}(T)}}$$

Asset Returns & Stylized Facts

Return Definitions

Financial econometric models analyze returns rather than raw prices due to stationarity and scale invariance.

Arithmetic vs. Logarithmic Returns

- **Simple (Arithmetic) Return:**

$$r_t = \frac{p_t - p_{t-1}}{p_{t-1}} \quad \Rightarrow \quad \text{Aggregates across assets in a portfolio.}$$

- **Logarithmic (Continuously Compounded) Return:**

$$R_t = \ln\left(\frac{p_t}{p_{t-1}}\right) \quad \Rightarrow \quad \text{Additive over time horizon } T.$$

The Bivariate Linear Regression Model

Model Formulation

Describes the linear dependency of a stochastic dependent variable y_t on an explanatory variable x_t:

$$y_t = \beta_0 + \beta_1 x_t + u_t, \quad t = 1, 2, \ldots, T$$

OLS Estimators Derivation

Minimizing the Residual Sum of Squares (RSS) $\sum \hat{u}_t^2$ yields normal equations:

$$\hat{\beta}_1 = \frac{\sum(x_t - \bar{x})(y_t - \bar{y})}{\sum(x_t - \bar{x})^2} = \frac{\mathrm{Cov}(x, y)}{\mathrm{Var}(x)}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

Standard Error of $\hat{\beta}_1$:

$$\mathrm{SE}(\hat{\beta}_1) = \frac{s}{\sqrt{\sum(x_t - \bar{x})^2}}, \quad \text{where } s = \sqrt{\frac{\sum \hat{u}_t^2}{T - 2}}$$

Gauss-Markov Assumptions (CLRM)

5 Core CLRM Assumptions

For OLS estimators to be **BLUE** (Best Linear Unbiased Estimator):

1. **Zero Mean Errors:** $E(u_t) = 0$.
2. **Homoskedasticity:** $\mathrm{Var}(u_t) = \sigma^2 < \infty$ for all t.
3. **No Autocorrelation:** $\mathrm{Cov}(u_i, u_j) = 0$ for all $i \neq j$.
4. **Exogeneity:** $\mathrm{Cov}(u_t, x_t) = 0$.
5. **Normality (Optional for Inference):** $u_t \sim N(0, \sigma^2)$.

Gauss-Markov Theorem

If assumptions 1–4 hold, OLS is the **Best Linear Unbiased Estimator (BLUE)**, achieving minimum variance among all linear unbiased estimators.

Statistical Inference & Hypothesis Testing

Testing Individual & Joint Parameters

Inference tests whether estimated relationships are statistically distinguishable from zero or hypothesized benchmark values.

Test Statistics Formulas

- **Individual *t*-test:** $H_0 : \beta_1 = \beta^*$

$$t_{\text{stat}} = \frac{\hat{\beta}_1 - \beta^*}{\mathrm{SE}(\hat{\beta}_1)} \sim t_{(T-k)}$$

- **Overall Goodness-of-Fit (R^2):**

$$R^2 = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\sum \hat{u}_t^2}{\sum (y_t - \bar{y})^2}$$

CLRM Diagnostic Testing & Violations

Diagnostic Matrix

Identifying and addressing econometric specification flaws.

Violations, Tests, and Remediation

Violation	**Detection Test**	**Consequence / Solution**
Heteroskedasticity	White Test, Goldfeld-Quandt	Wrong SEs; Use White Robust SEs
Autocorrelation	Durbin-Watson, Breusch-Godfrey	Biased SEs; Use Newey-West SEs
Non-Normality	Jarque-Bera (S, K)	Invalid t-tests in small samples
Multicollinearity	VIF > 10, Correlation Matrix	Unstable coefficients, large SEs
Misspecification	Ramsey RESET Test	Omitted variable bias; Respecify

The Capital Asset Pricing Model (CAPM)

Theoretical Foundation

The CAPM (Sharpe-Lintner-Mossin) formalizes the relationship between systematic risk (β) and expected return under the Efficient Market Hypothesis (EMH).

CAPM Equation & Two-Stage Empirical Formulation

- **Theoretical CAPM:**

$$E(R_i) - R_f = \beta_i \left[E(R_m) - R_f\right] \quad \text{where } \beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}$$

- **Stage 1 (Time-Series Regression):**

$$(R_{i,t} - R_{f,t}) = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + \epsilon_{i,t}$$

- **Stage 2 (Cross-Sectional Test):**

Fama-MacBeth (1973) Methodology

Methodological Innovation

Addresses non-constant betas and cross-sectional error correlation by estimating rolling time-series betas and running monthly cross-sectional regressions.

Fama-MacBeth Estimation Steps

1. **Step 1:** Estimate portfolio β_i over rolling 5-year windows ($t-60$ to $t-1$).
2. **Step 2:** Run cross-sectional regression for each month t:

$$R_{i,t} = \gamma_{0,t} + \gamma_{1,t}\hat{\beta}_{i,t-1} + z_{i,t}$$

3. **Step 3:** Calculate parameter averages and standard errors:

$$\hat{\lambda}_j = \frac{1}{T_{FMB}} \sum_{t=1}^{T_{FMB}} \hat{\gamma}_{j,t}, \quad t_{\text{stat}} = \frac{\hat{\lambda}_j}{\hat{\sigma}_\gamma / \sqrt{T_{FMB}}}$$

Fama-French Multi-Factor Models

Empirical Anomalies & Model Evolution

Market β alone fails to explain cross-sectional returns. Fama & French (1992, 1993, 1996) introduced factor additions to capture size and value effects.

Fama-French Factor Specification

- **Three-Factor Model (1993):**

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_i(R_{m,t} - R_{f,t}) + s_i \mathrm{SMB}_t + h_i \mathrm{HML}_t + e_{i,t}$$

  - SMB_t (Small Minus Big): Size risk premium.
  - HML_t (High Minus Low): Value risk premium (Book-to-Market).
- **Carhart Four-Factor Model (1997):** Adds UMD_t (Momentum / Up Minus Down).
- **Five-Factor Model (2015):** Adds RMW_t (Profitability) and CMA_t (Investment).

Stationarity & Autocorrelation

Weak (Covariance) Stationarity

A time series y_t is weakly stationary if its first two moments are finite and time-invariant:

1. $E(y_t) = \mu < \infty$
2. $\mathrm{Var}(y_t) = \gamma_0 < \infty$
3. $\mathrm{Cov}(y_t, y_{t-s}) = \gamma_s$ (depends only on lag s, not time t).

Autocorrelation Function (ACF)

The lag-s autocorrelation coefficient is defined as:

$$\tau_s = \frac{\gamma_s}{\gamma_0} = \frac{\mathrm{Cov}(y_t, y_{t-s})}{\mathrm{Var}(y_t)} \in [-1, +1]$$

For white noise ($y_t \sim WN(0, \sigma^2)$), the 95% confidence interval is $\pm 1.96/\sqrt{T}$.

Unit Root Testing & Spurious Regressions

The Non-Stationarity Problem

Regressing non-stationary series on each other causes **Spurious Regression** ($R^2 > d$, inflated t-stats with non-standard distributions).

Dickey-Fuller (DF) & Augmented Dickey-Fuller (ADF) Tests

Testing $H_0 : \phi = 1$ (Unit Root) vs $H_1 : |\phi| < 1$ (Stationary) in $y_t = \phi y_{t-1} + u_t$:

$$\Delta y_t = \psi y_{t-1} + \sum_{j=1}^{p} \alpha_j \Delta y_{t-j} + u_t \quad \text{(where } \psi = \phi - 1\text{)}$$

Test statistic $t_\psi = \frac{\hat{\psi}}{\mathrm{SE}(\hat{\psi})}$ follows Dickey-Fuller non-standard critical values.

- **PP Test:** Non-parametric correction for serial correlation.
- **KPSS Test:** Reverses hypotheses (H_0: Stationary).

ARMA Process & Box-Jenkins Methodology

ARMA(p, q) Specification

Combines Autoregressive $\mathrm{AR}(p)$ and Moving Average $\mathrm{MA}(q)$ components:

$$y_t = \mu + \sum_{i=1}^{p} \phi_i y_{t-i} + u_t + \sum_{j=1}^{q} \theta_j u_{t-j} \quad \Rightarrow \quad \phi(L) y_t = \mu + \theta(L) u_t$$

Box-Jenkins Model Identification Guide

Process	**ACF Plot**	**PACF Plot**
$\mathrm{AR}(p)$	Exponential / Sine-wave Decay	Cuts off sharply after lag p
$\mathrm{MA}(q)$	Cuts off sharply after lag q	Exponential / Sine-wave Decay
$\mathrm{ARMA}(p, q)$	Tail decay after lag q	Tail decay after lag p

Model Selection Criteria

Forecasting & Evaluation Metrics

Out-of-Sample Evaluation Philosophy

In-sample fit (R^2) often leads to overfitting. Forecasting performance must be tested out-of-sample on hold-out data.

Forecast Accuracy Measures

For N out-of-sample forecast errors $e_{t+s} = y_{t+s} - \hat{f}_{t,s}$:

- **Mean Squared Error (MSE):**

$$\mathrm{MSE} = \frac{1}{N}\sum_{t=1}^{N}\left(y_{t+s} - \hat{f}_{t,s}\right)^2$$

- **Mean Absolute Error (MAE):**

Summary & Methodological Roadmap

Key Takeaways

1. **Foundations:** Rigorous research requires understanding validity, measurement errors, and return properties.
2. **Regression Machinery:** CLRM diagnostic verification is vital before drawing inferential conclusions.
3. **Asset Pricing:** Multi-factor extensions (Fama-French) address systematic cross-sectional risk beyond CAPM β.
4. **Time Series:** Stationarity testing (ADF/KPSS) guards against spurious inferences in financial forecasting.

Next Step: Research Papers

- **Mid-term Paper:** Focus on Empirical Regression Diagnostics & Model Specification.
- **Final Paper:** Multi-Factor Asset Pricing Replication or ARMA Volatility Forecasting.